

[DC] Playful and participatory cultural heritage experience using extended reality technologies

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ABSTRACT

Creative design and applications based on Cultural Heritage (CH) content have gradually become a new way of communicating and interpreting cultural values. The proposed research aims to improve the CH experience using extended reality (XR) technologies by investigating the digital affordances of XR technologies that can support the design of playful CH experience, and exploring appropriate ways to enable user-generated content for participatory CH experience. This is an interdisciplinary project that sits at the intersection of human-computer interaction (HCI), XR, and CH. A practice-based approach will be adopted to conduct research in the wild, designing and developing novel XR interfaces and interaction techniques for CH experiences. The combined design thinking and computational approach will allow an in-depth understanding of playful and participatory CH experience using XR technologies, providing empirical, artifact, and theoretical contributions to the fields of HCI, XR, and CH.

Keywords: extended reality, cultural heritage, human-computer interaction

1 INTRODUCTION

Over the past decades, Cultural Heritage (CH) institutions have been working on conserving, preserving, and popularizing historical artifacts, monuments, sites, and intangible cultural heritage. The United Nations also included CH conservation in the Sustainable Development Goals in 2015. Increasing public interest in CH is beneficial to facilitate the development and implementation of policies that promote sustainable tourism, create job opportunities, and promote local culture [10]. Innovation in digital technologies is forming a new trend in CH visits. Some emerging technologies, such as Virtual Reality (VR) and Augmented Reality (AR), could allow people to travel between the real world and virtual worlds, and between the present and the past. These reality technologies that integrate virtual, augmented, and mixed reality are generally known as Extended Reality (XR) [9]. XR technologies have shown great potential for digital conservation, presentation, and interaction with cultural heritage [2, 6].

Owing to the evident strength of technological innovations, users are no longer merely satisfied with observing digital presentations of CH. Instead, they are inclined to seek immersive experiences that are more interactive and engaging. According to the review of papers published within five years related to cultural heritage, we learned that digital devices, interactive techniques, and multimedia allow for a connection between visitors and the digital content of CH, enhancing the visiting experience [13, 16]. Additionally, play, as an exploratory and experiential mode of engagement, is considered a key to inherently expanding and enhancing our embodied interaction with our surroundings [1]. Gameplay was also found to have positive

effects on CH learning and user engagement [8, 13]. Furthermore, the participatory culture, characterized by interaction, sharing, and common authorship [11], has been proven to engage visitors within the museum environment. This engagement can be further optimized by incorporating gameplay, thereby enhancing user experience [17].

However, some research gaps were found in previous work, including limited stakeholder involvement, lack of theoretical basis, inadequate evaluation studies, and missing design guidelines. These limit the ecological validity, reproducibility, and generalisability of the works. Thus, the proposed research aims to pursue the research agenda in XR and CH and fill the research gaps by taking a practice-based research approach. Specifically, we focus on playful and participatory CH experience.

2 OBJECTIVES AND RESEARCH QUESTIONS

2.1 Objectives

This research aims to use XR technologies to improve the CH experience. Specifically, we have three objectives:

- O1:** To understand the state-of-the-art approaches and stakeholders' attitudes towards XR technologies.
- O2:** To exploit the digital affordances and potential benefits of XR technologies in the CH experience.
- O3:** To investigate effective ways for users to actively participate in CH experience through XR technologies.

2.2 Research Questions

Many researchers and practitioners found XR technologies to bring opportunities for digital archiving, virtual reconstruction and visualization, and virtual curation of CH. However, there is a lack of understanding of the value that XR can provide to CH. Aside from visitors, the use of XR technologies in CH also concerns domain experts, curators, managers, policymakers, and other potential stakeholders. Establishing an understanding of stakeholder attitudes is crucial for subsequent research. Therefore, we ask **RQ1:** How do stakeholders perceive the state-of-the-art approaches of XR technologies in cultural heritage? Are there any cross-cultural differences?

Previous work showed that participants are highly engaged in exploration games in CH sites [7]. Some game mechanisms were found to have a positive effect on CH learning in aspects of motivation, effectiveness, and experience [8]. However, the design guidelines were unclear. Therefore, we propose **RQ2:** What are the digital affordances of XR technologies can benefit cultural heritage experience, and specifically, how can they support the design of playful cultural heritage experience?

In addition to the digital content formed in XR systems, user-generated content (UGC) will be the core component supporting the participatory CH experience. Towards the future metaverse, encouraging creativity and original content in the form of UGC is a direction worth putting effort into. This brings us to **RQ3:** What are the appropriate ways to support and moderate user-generated content for participatory cultural heritage experiences?

3 THEORETICAL FRAMEWORK AND METHODS

A practice-based approach will be adopted to conduct research in the wild (RITW). The proposed project will be in close collaboration with local and international partners. Within the field of HCI, RITW is becoming an increasingly popular approach. It typically involves the deployment of technology in a setting, using the methodology of ‘probing’ contexts, to change behavior or enhance community practice. The proposed research will follow Rogers and Marshall’s RITW framework [12] and refer to Sethu-Jones et al.’s explicit and principled rationale based on RITW [14]. This framework provides conceptual and practical guides to consider the different aspects involved in practice-based research, including the theory, design, technology, and in-situ studies.

Theory. We will use theories in HCI and CH when designing and developing XR systems. Some examples include the RITW framework [12], Double Diamond of Design [3], valence-arousal emotion model [5], Contextual Model of Learning [4], etc. By the end of the proposed research, we aim to summarise a theoretical framework that guides the design of playful and participatory CH experiences using XR technologies.

Design. We will adopt a participatory design approach [15]. Domain experts will be invited to interviews to obtain an in-depth understanding of their views. Potential target users and stakeholders will be involved to understand their requirements and expectations regarding the use of XR technologies in the CH experience through surveys or interviews. The results will inform the technology development.

Technology. This research focuses on using XR technologies, such as AR and VR. Thus, Head-Mounted Displays (HMDs) such as Meta Quest 2, HTC Vive Pro Eye, and Microsoft HoloLens are expected to be used. Mobile devices such as consumer-level smartphones are also likely to be used for AR development. XR systems will be developed in Unity with 3D models constructed and edited in Blender. Where necessary, photogrammetry techniques will be adopted to reconstruct photorealistic 3D assets. The research will lead to applications used in context, such as museums and galleries.

In-situ studies. We will conduct empirical in-situ studies to evaluate the design and refine the technology development. Specifically, a mixed method will be adopted to collect quantitative and qualitative data. Observations and field studies will be carried out, as these are typical HCI methods in natural settings. Where appropriate, the research seeks to record objective measures during the in-situ studies, such as user activity monitoring and electroencephalogram (EEG) monitoring. In addition, surveys and interviews will also be used to understand users’ subjective feelings, measure learning outcomes, and cross-examine the objective data analysis results.

4 CONCLUSION

There are three research questions that will be answered throughout the research process. Specifically, RQ1 will be studied based on existing work, supplemented by survey and interview; investigations of RQ2 and RQ3 will come with original designs. As a result of the practice-based research, it is expected that novel interfaces and interaction techniques, which are potentially patentable, will be developed. The proposed research will contribute to the knowledge base in HCI and CH.

The scientific value of the proposed work lies in the investigation of theoretical framework and design guidelines that can impact a wide range of interdisciplinary research in HCI, XR, and CH. In addition, ensuring the ecological validity of evaluation studies has long been an academic challenge in HCI research. The proposed work will address the challenge by conducting work in natural settings, using a practice-based research approach.

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